The Amphora Platform

Architecture, Technology Overview & Vision

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1. Executive Summary

Amphora Labs has built a manufacturing-first materials discovery platform that does something the "AI for materials" field does not: it produces candidates that are buildable, auditable, and supply-resilient — and it now improves itself.

The platform began as a neuro-symbolic architecture: neural models that generate and predict, governed by a symbolic layer of physics, chemistry, and manufacturing constraints that rejects anything unbuildable before a dollar is spent proving it. That architecture is running today and has already delivered a full program of results for our naval maritime coatings engagement with Huntington Ingalls Industries: a ten-candidate, zero-violation laboratory worklist scored across a fifteen-property target product profile, produced at a computational scale roughly one million times what a physical coatings laboratory tests in a year.

What is new — and what this document maps — is the layer we have built on top: a self-evolving system of autonomous agents and automated machine learning, an Amphora-owned capability architected by our Chief Technology Officer. Its mandate is simple to state and hard to copy: every learned model inside the platform is continuously evolved, benchmarked, and upgraded until it outperforms the best published alternative, with every upgrade adopted only on a measured win over frozen evaluation data. Our glass-transition-temperature prediction models — built as an early proof of concept, not a core deliverable — already exceed the published state of the art, and the same evolutionary machinery is now being applied across every model in the stack.

The result is a compounding system. The neuro-symbolic core makes discovery trustworthy. The compiler makes it manufacturable. The audit spine makes it verifiable by a skeptic. And the self-evolving layer makes the whole platform better every week, without waiting for the field to publish its next model — because when the field does publish, that model simply becomes another candidate our agents must beat.

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2. The Thesis: Discovery That Survives the Real World

Generative chemistry is commoditizing. Within a short horizon, producing a novel molecular structure with a theoretically attractive property will be an open-source commodity. Our operating thesis has been consistent from day one: **discovery has been solved; discovery that survives the move into the real world has not.**

What kills materials programs is not a shortage of interesting molecules. It is the translation gap: can it be made from commercially available, non-restricted feedstocks? Does it mix at the ratio a shipyard technician's spray rig is calibrated for? Does it cure in the pot-life window? Does it survive the supply-chain shock when a precursor disappears? Can a customer's counsel and chemists audit the claim trail rather than trust a vendor's slide?

Amphora's platform is engineered around that gap. Manufacturability is not a filter applied after discovery — it is a constraint compiled into generation itself. Evidence status is not a caveat in a footnote — it is part of each candidate's machine state, and missing evidence fails closed rather than being imputed as a pass. This is why our selection yield runs 2.18 times higher than the conventional screen-then-discard approach under the same laboratory budget, and why our deliverables survive hostile technical review.

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3. The Architecture at a Glance

The platform is a layered stack with two cross-cutting systems. Read bottom to top:

Layer	Name	Function	Status
0	Data & Knowledge Foundation	Public corpora, proprietary datasets, live supply and patent feeds	Running
1	Neuro-Symbolic Core	Neural generation and prediction, governed by symbolic physics, chemistry, and manufacturing constraints, joined by a knowledge graph	Running; knowledge graph in active build
2	Manufacturing Compiler	Deterministic pipeline: generate → score → filter for manufacturability → rank → audit → hand off	Running
3	Programs & Deliverables	Vertical applications; naval maritime coatings is the flagship	Running (coatings); next verticals staged
Cross-cutting	Trust & Audit Spine	Frozen instruments, fail-closed evidence handling, tamper-evident receipts, independent grading	Running
Cross-cutting	Self-Evolving Layer	Autonomous agents plus automated machine learning that continuously improves every model, pipeline, and dataset in the stack	Running (v1); expanding across the full stack

The two cross-cutting systems are in deliberate tension, and that tension is the design's core insight: **the platform evolves, but it cannot grade itself.** Generators, predictors, and pipelines improve continuously under the self-evolving layer; the judges, gates, and audit instruments they are measured against stay frozen within each release. A self-improving system whose evaluators are frozen produces genuinely better models. A self-improving system that can edit its own evaluator produces better-looking numbers. We built the former.

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4. Layer 0 — Data & Knowledge Foundation

Everything above this layer is only as good as what feeds it. The foundation combines three classes of data:

Public scientific corpora. The full PI1M polymer informatics corpus (988,775 polymers), the PolyMetriX glass-transition dataset (7,367 measured records), the Open Macromolecular Genome commercial monomer library (48,213 structures), RadonPy

molecular-dynamics density data (1,077 systems), and a curated set of 51 literature-cited, experimentally reported two-dimensional building blocks spanning seven covalent organic framework linkage families. These power training, benchmarking, and novelty screening.

Proprietary Amphora assets. The most strategically important is our cured-network glass-transition dataset: experimental anchors from primary literature paired with an at-scale molecular-dynamics dataset of cured thermoset systems, and growing. Nearly all public glass-transition predictors operate on linear repeat units; the property that actually matters for a coating is the cured, crosslinked network. This dataset is held as a trade secret and is the seed of a durable data moat. Alongside it: the feedstock and process graph (role classes, equivalent weights, source evidence, process windows), program-generated candidate and evidence ledgers, and reverse-engineering data from destructive testing of incumbent materials.

Live external feeds. Patent-chemistry databases (PubChem and SureChEMBL, queried live — our current leads were verified exact-novel against 116,894 real patents referencing the precursor genus), vendor catalogs, and pricing and availability interfaces for supply-chain routing.

The self-evolving layer's data agents (Section 8) treat this foundation as territory to expand: sourcing new corpora, curating program data into training assets, and flagging coverage gaps before they become model blind spots.

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5. Layer 1 — The Neuro-Symbolic Core

This is the base of the architecture and the origin of the company. It has two engines joined by a knowledge graph.

5.1 The Neural Engine

The learning half of the core. Current production components:

Generation. A trained character-level recurrent generator, retrained on the full polymer corpus at 92.8% structural validity, which delivered a measured 10.2× structural-diversity gain over our earlier rule-based search (163 versus 16 structural clusters at matched sample size). A conditional diffusion generator is scoped as the next-generation upgrade. Generation also includes fragment-recombination operators and a bioisosteric "escape" operator that hops a candidate out of a patented structural genus while ranking each hop by attorney-defensibility.

Recognition. A learned chemistry recognizer (gradient-boosted trees) validated under a leave-one-family-out protocol against functional-group-matched hard decoys — the hardened benchmark where our recognizer scores a contestable 0.969 area-under-curve while the trivial similarity baseline collapses to 0.78. This model, not similarity-to-seeds, is

the headline discovery objective: candidates are judged by generalization to chemistry the model has never seen. In blind testing the engine surfaced elite published chemistry — including the two-dimensional polyaramid monomer from the landmark Nature 2022 paper — unprompted, against decoys.

Property prediction. Stacked ensembles of gradient-boosted tree models over molecular fingerprints and descriptors, with a polymer language-model embedding head. Our glass-transition-temperature models — originally a proof of concept rather than a coatings deliverable — now exceed the published state of the art (internal benchmark result; publication-grade receipt scheduled for the next white-paper revision). The stack also includes a measured out-of-distribution upgrade: a tabular foundation model (TabPFN-v2) that won our four-way bake-off on scaffold-split data, now deployed behind a learned novelty router that sends in-distribution queries to the tree ensemble and out-of-distribution queries to the foundation model. This router is the template for how the self-evolving layer adopts winners: measured, split-frozen, and routed — never asserted.

Continuous search. Polymer language-model embeddings power a continuous similarity search alongside the discrete kernel; the two disagree on nearest neighbors 57% of the time, which is exactly why we run both.

5.2 The Symbolic Engine

The governing half — the reason our output is buildable rather than merely interesting.

The physics judge bank. Literature-anchored, auditable physical models scoring every candidate across the full target product profile: network glass-transition (DiBenedetto relation), cure kinetics (Kamal-Sourour, modulated by quantum-chemistry electronic signals), work of adhesion (Owens-Wendt), fractional free volume as a barrier proxy (Bondi relation), solubility (Fedors group contribution, validated to roughly 4% on solvents), and molecular-dynamics simulation (OpenFF force fields in OpenMM, 150-copy cells, 100,000-step constant-pressure runs) on gated top candidates. These are physics-grounded proxies, honestly bounded — auditable, not a black box.

Fail-closed evidence gates. Every candidate carries an explicit gate-state vector: pass, fail, or unknown-fails-closed. Missing measured data can never be imputed as a pass. In the maritime coatings program this discipline preserved twenty unknown gates as visible measurement obligations instead of converting them into claims — which is precisely what a sophisticated customer's diligence team checks first.

Coupled two-part projection. The compiler's signature move for thermoset systems: rather than optimizing a single molecule, it projects candidates into paired Part-A/Part-B formulation hypotheses with stoichiometric equivalent closure — enforcing, for example, the one-to-one by-volume mix ratio that lets a shipyard use existing dual-feed spray equipment without recalibration.

Exact similarity search at superhuman scale. A custom graphics-card kernel computes exact Tanimoto similarity (the standard fingerprint-overlap measure) at 5.71 billion comparisons per second on a single consumer card — bit-exact against an independent reference implementation, screening the entire public polymer corpus for novelty in 1.55 seconds. This is the search half of "beyond human": a measured wall-clock number, not a projection.

Synthesis and legal instruments. A synthesis-readiness classifier (S0–S4) with route cards for every promoted candidate, and a freedom-to-operate instrument grounded in live patent queries with automated claim-charting — evidence for counsel, never a legal conclusion.

5.3 The Knowledge Graph

The seam joining the two engines: a graph of materials, feedstocks, roles, equivalent weights, process windows, evidence provenance, and program outcomes. Today it exists as the feedstock/process graph driving the compiler; it is being built out into the full neuro-symbolic knowledge graph as every program cycle — successes and failures alike — writes structured relational facts back into it. This is the asset that compresses each successive program: the graph a new vertical inherits is the accumulated, machine-readable manufacturing reality of every vertical before it.

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6. Layer 2 — The Manufacturing Compiler

The core pipeline is deterministic and fixed: **generate → score → filter for manufacturability → rank → audit → hand off.** From a target specification, the engine novelty-screens on the order of ten billion candidate molecules per day at cluster scale, scores buildable two-part formulations across all fifteen modeled properties at roughly 55 million per day on a single processor core, and returns a vetted, supply-resilient shortlist in hours. For calibration: a physical coatings laboratory tests on the order of fifty formulations per year. That is roughly a million-fold scale advantage in candidates scored per unit time — with the essential caveat, always stated, that scored means computationally evaluated across all properties, not laboratory-tested.

Two compiler capabilities deserve executive attention:

Full-profile scoring, not hero properties. Incumbent approaches optimize one headline property and hope the rest follows. The compiler scores every candidate against the entire target product profile — anti-corrosion, anti-fouling, mechanical, thermal and chemical durability, and application constraints — simultaneously. This is how it surfaces non-obvious winners a single-property selector would rank low.

Supply-resilient re-compilation. When a precursor is lost — discontinued, sanctioned, priced out — the engine re-compiles a buildable substitute that preserves the modeled performance envelope. In the coatings program, losing the lead hardener re-compiles to the alternate while preserving 100% of the key-axis envelope. A supply-chain shock never restarts the clock. This capability is a product in its own right and a wedge into every manufacturer living with brittle chemical supply chains.

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7. The Trust & Audit Spine

This spine runs through every layer and is, in our considered view, the deepest moat in the company — because a competitor can re-implement an algorithm, but cannot retroactively make their claims re-derivable.

Its mechanisms: every scoring instrument is **frozen** per release — a pure function of the input structure, so any skeptic can re-derive any score and get the same number. Every reported value carries a **boundary mode** in a strict lattice (fixture → locally generated → web-grounded → modeled → production), and production-grade claims — cost per lead, wet-lab performance, procurement proof — **fail closed** when no evidence exists. An **independent grader of a different model class** (deterministic cheminformatics, no language model, no shared weights with the generator) re-derives the proximity, synthesis-class, and patent claims of every promoted candidate, and is exercised by an adversarial tamper battery of injected cheats — twelve out of twelve caught. The promotion signal for the self-improving loop is a **set-difference against a sealed reference ledger** — a deterministic integer the system cannot inflate by re-grading itself. And every shipped package carries a **hash-chained, cryptographically signed audit receipt** with a one-command verification script that re-derives every headline number from saved artifacts.

For investors, the commercial translation: our deliverable to an enterprise partner is not a ranked list, it is a system of record their counsel and chemists can audit instead of trust. That is what converts a pilot into a platform relationship.

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8. Layer 3 (Cross-Cutting) — The Self-Evolving Layer

This is the newest and fastest-compounding part of the platform: an Amphora-owned automated machine learning and autonomous-agent capability, architected by our Chief Technology Officer, deployed as an internal tool across the entire stack.

8.1 What it does

Automated machine learning with a benchmark-beating mandate. The layer continuously searches over model architectures, features, and training regimes for every

learned model in the platform — generators, recognizers, property predictors, surrogates, routers — trains challengers, and evaluates them against both our incumbent production models and the best published external benchmarks on frozen data splits. Its design mandate: **continuously improve every internal model until it outperforms all benchmarks it targets, and keep it there as the field moves.** Adoption is never asserted; a challenger is promoted only on a measured, confidence-separated win, and the promotion is written into the audit ledger like any other claim. Our glass-transition models exceeding the published state of the art is the first proof of the mandate; the tabular foundation-model router was the first proof of the adoption discipline.

Autonomous agents across the stack. Agents operate at every layer: optimizing pipeline stages and compute allocation; inventing and testing new model candidates; sourcing, curating, and managing datasets into the foundation layer; monitoring benchmark drift as new external models publish; and maintaining the evidence ledgers. The agents run inside sandboxed tool environments partitioned from controlled-data repositories, so autonomy never crosses a compliance boundary.

The meta-evolver loop. A self-improving search process whose reward is the ungameable set-difference signal described in Section 7: an agent-driven system earns credit only for genuinely novel, genuinely promoted discoveries against a sealed ledger. This is what let us diagnose and fix generator mode-collapse with measured evidence (the 10.2× diversity recovery) rather than intuition.

8.2 Why this is defensible

Three reasons. First, the layer inherits the trust spine: because evaluators are frozen and promotions are ledgered, its self-improvement is provable rather than promotional — most automated machine learning claims in the market are neither. Second, it converts the field's progress into our progress: every new state-of-the-art publication becomes a challenger our agents must beat or absorb, which is the anti-obsolescence property in its strongest form. Third, it compounds on proprietary data the field does not have — the cured-network dataset, the program evidence ledgers, the knowledge graph — so parity on public benchmarks understates our advantage on the problems customers actually pay for.

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9. Proof: The Naval Maritime Coatings Program

The flagship program, with Huntington Ingalls Industries, targets the Navy's military specification for ship coatings (MIL-DTL-24441). It is both a revenue engagement and the platform's first end-to-end demonstration. Methods and shipped results:

Method stack used. Coupled Part-A/Part-B projection over source-screened commercial feedstocks at a one-to-one by-volume mix; fail-closed gating; fifteen-property judge-bank scoring of the full target product profile; cured-network glass-transition modeling

(molecular dynamics plus literature anchors); molecular-dynamics physics gates on top leads; exact full-corpus novelty screening; live patent grounding; supply-resilient re-compilation; and the complete audit spine on every claim.

Shipped results. A coupled two-part laboratory worklist of ten executable hypotheses with zero hard-gate violations and twenty unknown gates preserved fail-closed. A 2.18× constraint-yield advantage over post-hoc screening under the same laboratory budget. A recommended two-layer coating system — a high-glass-transition barrier primer with a companion anti-fouling topcoat track — predicted at target or better on eleven of thirteen primer-owned properties, including a roughly 150 °C glass transition a conventional polyamide system cannot reach, at low volatile-organic-compound content and a field-simple mix. A funded, phased wet-lab gate plan that puts the cheapest disqualifying measurements first. Every performance value is modeled and honestly bounded pending laboratory measurement — and that discipline is itself part of what we sold.

What the program feeds back. Every measured result — pass or fail — recalibrates the judge bank, extends the cured-network dataset, and writes structured facts into the knowledge graph. The customer's laboratory spend sharpens our platform; that is the co-development flywheel in miniature.

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10. Roadmap: Now, Next, Horizon

Now (running, 2026). The manufacturing compiler, neuro-symbolic core, and trust spine in production. The maritime coatings program delivered through its computational phase, wet-lab gate plan staged. Self-evolving layer version one live: meta-evolver loop, trained generator, continuous benchmarking harness, first benchmark-exceeding models and first measured-upgrade adoptions. Cured-network dataset growing as a trade-secret asset.

Next (in active build, 2026–2027). The self-evolving layer extends across the entire model stack — every property model, surrogate, and pipeline stage under continuous agent-driven evolution against published benchmarks. The knowledge graph is constructed at full scope from accumulating program data, shifting search from brute-force sweeps toward graph-guided inference. Wet-lab results from the coatings gate plan close the loop, promoting calibrated physics gates from soft screens to hard gates with quantified error. Agents assume end-to-end pipeline operation, with human sign-off retained on every safety-critical and release decision. Adjacent polymer and coatings programs onboard onto the compiled platform at a fraction of first-program cost.

Horizon (2027 and beyond). The platform generalizes beyond organic coatings into adjacent materials classes and extreme-environment applications, carried by the same kernel: the compiler, the trust spine, and the self-evolving layer are materials-agnostic by construction, with domain physics and datasets added as plugins. Supply-resilient re-

compilation matures into a standalone product for any manufacturer exposed to chemical supply shocks or regulatory bans. The end state: Amphora as the system of record for manufacturable materials evidence — the place where a claim about a material's buildability, performance, and provenance is created, verified, and trusted — with an autonomous discovery-to-pilot loop running under human governance.

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11. Why This Wins

Five reinforcing moats, in order of durability:

The fail-closed system of record. Re-derivable claims, frozen instruments, tamper-evident receipts. Code can be re-implemented; a trust architecture and the customer relationships built on it cannot be copied by re-implementing an algorithm.

Proprietary data that compounds. The cured-network dataset, program evidence ledgers, and the knowledge graph grow with every engagement, and customer-funded measurement makes them grow faster — the services-as-wedge dynamic working for the platform, not against it.

Manufacturing-first constraints. The coupled two-part projection, stoichiometric closure, sourceability gating, and re-compilation are built from industrial reality that generic discovery platforms have not encoded and will take years to learn.

The self-evolving layer. A provable, ledgered self-improvement capability with a mandate to outperform every benchmark — which turns the field's publication velocity from a threat into an input.

Anti-obsolescence by architecture. Any new state-of-the-art generative or predictive model becomes a plugin feeding our established brain and judges. We do not bet against progress in machine learning; we are structured to consume it.

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12. Evidence Discipline (Claims Policy)

This document is internal and forward-looking, and it follows the same discipline as our external materials: results stated here as measured are measured and re-derivable from ledgered artifacts; results stated as internal benchmarks carry a scheduled publication receipt; forward statements (the benchmark mandate, the roadmap) are design commitments verified continuously by the benchmarking harness, not present-tense performance claims. Nothing in this document asserts wet-laboratory coating performance, qualification acceptance, freedom-to-operate, or procurement readiness; those claims unlock only on their receipts. This discipline is not caution — it is the product.
